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NETWORK TRACKING ANALYSIS USING PCAP NETWORK TRAFFIC ANALYZER

CSE23CL301 - Computer Networks Laboratory

PROJECT REPORT

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BONAFIDE CERTIFICATE

Certified that this project report "Network Tracking Analysis Using PCAP Network Traffic Analyzer" is the bonafide record of work done by "A Surya Kumar - E0324079, M Kishore Kumar - E0324081 and J R Nikilesh Gajan - E0324098" who carried out the project work under my supervision as part of the course "CSE23CL301 - Computer Networks Laboratory" in partial fulfillment of the requirements for the award of the degree in Computer Science and Engineering during the academic year 2026-2027.

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ABSTRACT

Network administrators read packet captures to find out what is happening on a network, but tools such as Wireshark show individual packets and leave the analyst to spot patterns and attacks by hand. This project builds the PCAP Network Traffic Analyzer, a web application written with Streamlit that reads a packet capture (PCAP or PCAPNG, up to 200MB) and turns it into summaries, charts and alerts. A home page reports the number of packets, the capture duration, the protocol mix and the packets sent per second; separate pages cover TCP handshakes, DNS, HTTP, LAN traffic and quality of service. An alerts page combines three rule-based detectors, for SYN floods, port scans and DNS tunnelling in 10-second windows, with an Isolation Forest anomaly score for each source IP address. We tested the application on a synthetic capture of 170 packets over 4.70s (162 TCP and 8 DNS packets) that contains one example of each attack. The analyzer flagged all three sources: 10.0.0.5 as a SYN flood (130 SYNs with a low SYN-ACK ratio, risk score 70), 10.0.0.9 as a port scan (32 destination ports, risk score and 10.0.0.7 as DNS tunnelling (risk score 100). All 162 TCP flows were reported as half-open, with no retransmissions or duplicate ACKs. The Isolation Forest agreed with the rules on two of the three sources but gave the port scanner a score of 0, which shows the limits of unsupervised scoring on a capture with only three hosts. The results come from one small synthetic capture, so detection accuracy on real traffic has not yet been measured. The tool is intended for teaching and laboratory use.

Keywords: network traffic analysis, packet capture, PCAP, TCP handshake, SYN flood, port scan, DNS tunnelling, Isolation Forest, anomaly detection, Streamlit.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
ACK	Acknowledgement (TCP flag)
DNS	Domain Name System
DoS	Denial of Service
HTTP	Hypertext Transfer Protocol
ICMP	Internet Control Message Protocol
IDS	Intrusion Detection System
IETF	Internet Engineering Task Force
IP	Internet Protocol
LAN	Local Area Network
ML	Machine Learning
NAT	Network Address Translation
PCAP	Packet Capture (file format)
PCAPNG	PCAP Next Generation (file format)
QoS	Quality of Service
RFC	Request for Comments
SYN	Synchronise (TCP flag)
SYN-ACK	Synchronise-Acknowledgement (second step of the TCP handshake)
TCP	Transmission Control Protocol
UDP	User Datagram Protocol
UI	User Interface

CHAPTER 1

INTRODUCTION

1.1 Background

Every message that crosses a computer network is carried in packets, and a packet capture records those packets with their headers and timestamps. Captures are saved in the PCAP format used by tcpdump and libpcap [3] or in its successor PCAPNG, and they are the main evidence an administrator has when a link is slow, a service fails or an attack is suspected.

Reading a capture packet by packet is slow. A short capture can hold thousands of packets, and the patterns that matter, such as a host that opens many connections and never completes them, appear only when packets are grouped by host, by flow and by time. Three such patterns are common in teaching and in practice. A SYN flood sends many TCP connection requests without completing the handshake, so the server holds many half-open connections. A port scan probes many destination ports of a host to find open services. DNS tunnelling hides data inside DNS queries, which produces long or random-looking domain labels.

Rule-based detectors catch these patterns when they cross a threshold, but they miss behaviour no rule describes. Unsupervised anomaly detection, such as the Isolation Forest [7], can rank hosts by how unusual they are without labelled examples. This project combines both in one application that a student or administrator can run in a browser.

1.2 Problem Statement

Given a packet capture, the system must (i) summarise the traffic: the number of packets, the duration, the protocol mix and the packet rate over time; (ii) analyse TCP connections, including the state of each three-way handshake, retransmissions and duplicate acknowledgements; (iii) present DNS, HTTP, LAN and quality-of-service views of the same capture; (iv) detect SYN floods, port scans and DNS tunnelling and explain each finding in plain words; and (v) give each source IP address a severity, a risk score and an anomaly score, all in a browser without command-line tools.

1.3 Objectives

- To build a web application that accepts PCAP and PCAPNG captures of up to 200MB, or a built-in sample, and parses every packet.

- To show an overview of each capture: packets, duration, number of protocols, protocol breakdown and packets per second.
- To analyse TCP flows by handshake state and to count retransmissions and duplicate ACKs.
- To provide separate pages for DNS, HTTP, LAN and QoS analysis.
- To detect SYN floods, port scans and DNS tunnelling with rules over 10-second windows, and to state the reason for each alert.
- To score each source IP with an Isolation Forest and combine the result with the rules into a severity and a risk score.
- To test the application on a synthetic capture that contains one example of each attack.

1.4 Approach

The application reads a capture, extracts the header fields of each packet and builds tables of packets, flows and source hosts. Summary statistics and charts are computed from these tables, three detectors examine each source in 10-second windows, and an Isolation Forest scores each source IP. The results are shown on seven pages of a Streamlit application: the home page (app) and the TCP, DNS, HTTP, LAN, QoS and Alerts pages.

1.4.1 Justification of the approach

Why a web application. A browser page needs no installation by the user and shows charts, tables and alerts together. Streamlit turns a Python script into such a page with little extra code, and Plotly gives interactive charts that can be zoomed and saved.

Why rules for the known attacks. SYN floods, port scans and DNS tunnelling have clear signatures: many SYNs with few SYN-ACKs, many destination ports from one source, and long or random-looking DNS labels [4]–[6]. A rule can state its reason in one sentence, which an alert must do to be useful.

Why an Isolation Forest as well. Rules only find what they describe. An Isolation Forest needs no labelled attacks and isolates unusual points in few random splits, so it can point at hosts that no rule flags. Its score is shown next to the rule result, not instead of it.

Why a synthetic capture. A capture built to contain known attacks gives a ground truth: we know which host did what, so we can check whether each detector fired. Real captures with labelled attacks are hard to obtain and may hold private data.

1.5 Scope and Organisation

Chapter 2 describes the captures used. Chapter 3 describes how packets are prepared for analysis. Chapter 4 explains the analysis methods, the detection rules, the Isolation Forest and the evaluation measures. Chapter 5 describes the implementation and Chapter 6 the results on the synthetic capture, with screenshots of the running application. Chapter 7 discusses the findings, limitations and future work, and Chapter 8 concludes.

The analyzer is a teaching and laboratory tool. It is not a production intrusion detection system: it analyses saved captures, not live traffic, and its detectors have been tested only on one small synthetic capture.

CHAPTER 2

DATASET DESCRIPTION

The application analyses packet captures. Two sources of captures are supported: files that the user uploads and sample captures built into the application.

2.1 Uploaded Captures

The user can upload a capture in PCAP or PCAPNG format of up to 200MB per file (Figure 6.1). Any capture saved by Wireshark or tcpdump can be used.

2.2 Synthetic Attack Capture

The sample synthetic_attacks.pcap was used for all results in this report. It holds 170 packets over 4.70s from three source hosts on the 10.0.0.0/24 network. Each host shows one kind of attack: 10.0.0.5 sends 130 TCP SYN packets (a SYN flood), 10.0.0.9 sends 32 TCP packets to 32 different destination ports (a port scan), and 10.0.0.7 sends 8 DNS queries with long or random-looking labels (DNS tunnelling).

2.3 Summary

Table 2.1 summarises the synthetic capture. The per-host packet counts are those reported by the application (Table 6.4). Because every host in this capture is an attacker, the capture has no normal traffic. This matters for the Isolation Forest, which looks for hosts that differ from the rest (Section 6.6).

Table 2.1: Dataset summary: synthetic_attacks.pcap.

Property	Value
File format	PCAP
Packets	170
Duration	4.70s
Protocols	2 (TCP 162 packets, DNS 8 packets)
Source hosts	3 (10.0.0.5, 10.0.0.7, 10.0.0.9)
10.0.0.5	130 packets, 7,020 bytes – SYN flood
10.0.0.9	32 packets, 1,728 bytes – port scan
10.0.0.7	8 packets, 968 bytes – DNS tunnelling
Labels	One known attack per source host

CHAPTER 3

DATA PREPROCESSING

Before analysis each capture passes through the steps below, in order.

3.1 Packet Parsing

The capture file is read packet by packet. For each packet the timestamp, length, source and destination IP addresses, transport protocol, ports and TCP flags are extracted; for DNS packets the queried name is also kept.

3.2 Protocol Labelling

Each packet is labelled with the highest protocol that can be recognised, so a UDP packet to port 53 is counted as DNS rather than UDP. In the synthetic capture this gives two protocols, TCP and DNS (Figure 6.2).

3.3 Time Binning

Timestamps are shifted so that the first packet is at time zero and then grouped into one-second bins for the packets-over-time chart. For the detection rules the packets of each source are grouped into 10-second windows, which is the window named in the alert reasons (Figure 6.4).

3.4 Flow Construction

TCP packets are grouped into flows by source address, source port, destination address and destination port. For each flow the application records which steps of the three-way handshake were seen (SYN, SYN-ACK, ACK), so each flow can be labelled as complete or half-open (Figure 6.3).

3.5 Per-Host Features

For each source IP the application counts packets and bytes, which appear in the toptalkers table (Figure 6.5). These per-host statistics form the feature vector that the Isolation Forest scores, one vector per source IP.

CHAPTER 4

METHODOLOGY

4.1 Traffic Overview

The overview reports the number of packets N, the capture duration and the number of distinct protocols. The duration is

$$D = t_{\text{last}} - t_{\text{first}}, \tag{4.1}$$

where t_{first} and t_{last} are the timestamps of the first and last packets. The protocol breakdown is the share of packets of each protocol, and the packet rate is the number of packets in each one-second bin.

4.2 TCP Handshake Analysis

A TCP connection opens with a three-way handshake: the client sends SYN, the server answers SYN-ACK and the client completes it with ACK [4]. A flow in which the SYN was seen but the handshake never completed is half-open. The TCP page also counts retransmissions (a segment sent again with a sequence number already seen) and duplicate ACKs (the same acknowledgement number received more than once), which indicate loss on the path.

4.3 SYN Flood Rule

A SYN flood leaves many half-open connections on the target [5]. For each source and 10-second window the rule counts SYN packets and computes

$$r = \frac{N_{\text{SYN-ACK}}}{N_{\text{SYN}}}, \tag{4.2}$$

where N_{SYN} is the number of SYNs sent by the source and $N_{\text{SYN-ACK}}$ the number of SYN-ACKs returned to it. The rule fires when N_{SYN} is high and r is low within one window. A SYN flood finding carries 70 severity points (Table 6.3).

4.4 Port Scan Rule

A port scan contacts many destination ports in a short time. For each source and 10second window the rule counts the distinct destination ports P and fires when P reaches a set threshold. A port scan finding carries 45 severity points.

4.5 DNS Tunnelling Rule

Data hidden in DNS queries makes the labels of the queried names long and close to random. A common measure of this randomness is the Shannon entropy of the characters of a label:

$$H = -\sum_{i} p_i \log_2 p_i, \tag{4.3}$$

where p_i is the relative frequency of character i in the label. The rule fires when a source queries labels that are unusually long or random-looking. A DNS tunnelling finding carries 55 severity points.

4.6 Isolation Forest Scoring

An Isolation Forest builds many random trees, each of which splits the data on a random feature at a random value until every point is isolated. Anomalous points are isolated in fewer splits. The anomaly score of a point x among n points is

$$s(x,n) = 2^{-E[h(x)]/c(n)}, \tag{4.4}$$

where $E[h(x)]$ is the mean path length of x over the trees and $c(n)$ is the mean path length of an unsuccessful search in a binary search tree of n points; s close to 1 marks an anomaly. The application fits the forest on the per-host features of the same capture ("same-capture exploratory scoring") and reports the score on a 0–100 scale as ml_anomaly_score. A capture can instead be marked as a normal baseline, so that later captures are scored against it.

4.7 Severity and Risk Score

Each source receives a risk score from 0 to 100 and a severity of Low, Medium or High. In the results the risk score equals the rule points for 10.0.0.5 (70) and 10.0.0.9 (45), and rises to 100 for 10.0.0.7, whose anomaly score is also 100 (Table 6.2).

4.8 Evaluation Metrics

Because the synthetic capture has one known attack per host, each detector is evaluated by whether it fired for the right host and stayed silent for the others. With TP the number of attacking hosts correctly flagged and FN the number missed, the detection rate is

$$DR = \frac{TP}{TP+FN} \times 100\%. \tag{4.5}$$

False alerts are counted as findings that name the wrong attack for a host. The traffic statistics are checked for internal consistency, for example that per-protocol counts sum to the total number of packets and that per-host bytes divided by packets give plausible packet sizes.

CHAPTER 5

IMPLEMENTATION

5.1 System Architecture

Figure 5.1 shows the structure of the application. A capture enters either by upload or as a built-in sample; the parser extracts the packet fields; the overview, the protocol analyses, the rule-based detectors and the Isolation Forest work on the parsed packets; and the results are shown on the pages of the Streamlit application.

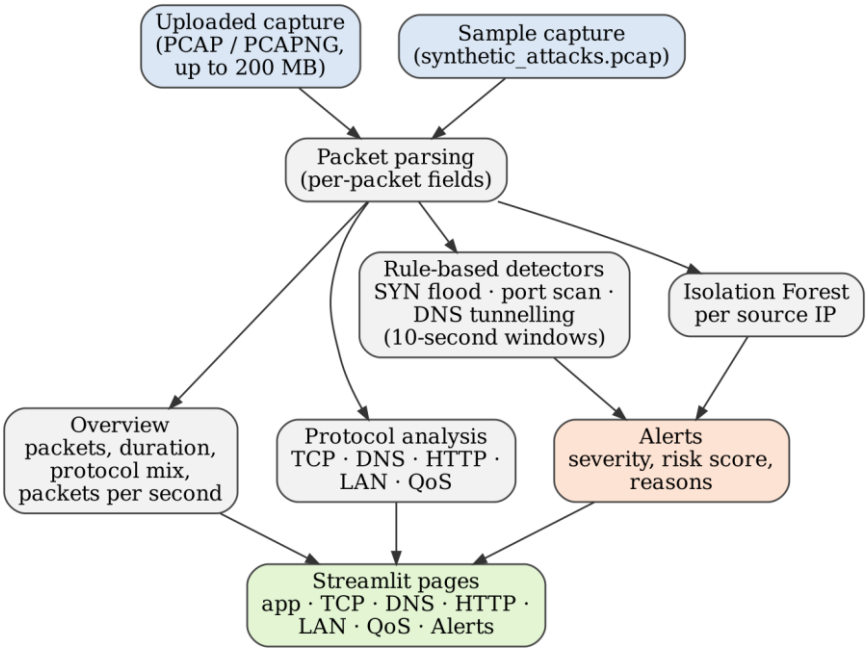


Figure 5.1: Overview of the PCAP Network Traffic Analyzer. A capture is parsed once, and its packets feed the overview, the protocol pages, the rule-based detectors and the Isolation Forest, whose results appear on the Streamlit pages.

5.2 Application Pages

The application has seven pages, listed in the sidebar (Figure 6.1). Table 5.1 describes each page; the footer of the home page lists the laboratory topics they cover: TCP handshake, DNS, HTTP, LAN/NAT, reliability and QoS.

Table 5.1: Pages of the application.

Page	Content
app (home)	Upload or sample selection, Analyze capture button, packets, duration, protocols, protocol breakdown, packets over time, top talkers
TCP	TCP 3-way-handshake lab: TCP flows, retransmissions, duplicate ACKs, flow status by handshake state
DNS	DNS lab: the DNS queries in the capture
HTTP	HTTP lab: the HTTP traffic in the capture
LAN	LAN/NAT lab: local-network and address-translation view of the capture
QoS	QoS lab: quality-of-service and reliability view of the capture
Alerts	Rule-based findings and Isolation Forest score per source IP, severity filter, option to mark a capture as normal baseline

5.3 Capture Input

The home page offers a file uploader for PCAP and PCAPNG files of up to 200MB and a drop-down list of sample captures. Pressing *Analyze capture* parses the chosen capture and reports the number of packets analysed; the results are then available on every page of the sidebar.

5.4 Charts and Tables

Charts are drawn with Plotly, so each can be zoomed, panned and saved as an image from its toolbar. The protocol breakdown is a pie chart, packets over time and flow status are line and bar charts, and the alerts and top talkers are interactive tables.

5.5 Alerts and Baseline

The Alerts page shows one row per flagged source with its severity, risk score, anomaly score and reasons, and a second table with one row per rule finding (Figure 6.4). A multi-select filter limits the view to High, Medium or Low alerts. The button *Mark this capture as normal baseline* stores the current capture as normal traffic, so that the Isolation Forest can score later captures against it instead of against themselves.

5.6 Setup and Settings

Table 5.2 lists the software used and Table 5.3 the main settings visible in the application.

Table 5.2: Software Settings

Item	Use
Python	Programming language of the application
Streamlit	Web application and multi-page layout [8]
Plotly	Interactive charts [9]
Isolation Forest	Unsupervised anomaly scoring [7]
PCAP / PCAPNG	Capture file formats read by the application [3]
Web browser	Application served locally at localhost:8501

Table 5.3: Main Settings

Setting	Value
Accepted formats	PCAP, PCAPNG
Maximum upload size	200MB per file
Packet-rate bin	1 second
Detection window	10 seconds
Rule points	SYN flood 70, DNS tunnelling 55, port scan 45
Severity levels	High, Medium, Low
ML mode	Same-capture exploratory scoring, or scoring against a normal baseline

CHAPTER 6

RESULTS

This chapter reports the outputs of the application on the synthetic capture synthetic_attacks.pcap. Every number below was read from the running application, and every figure is a screenshot of it.

6.1 Capture Overview

Figure 6.1 shows the home page after the sample was analysed. The application analysed 170 packets over 4.70s and found 2 protocols.

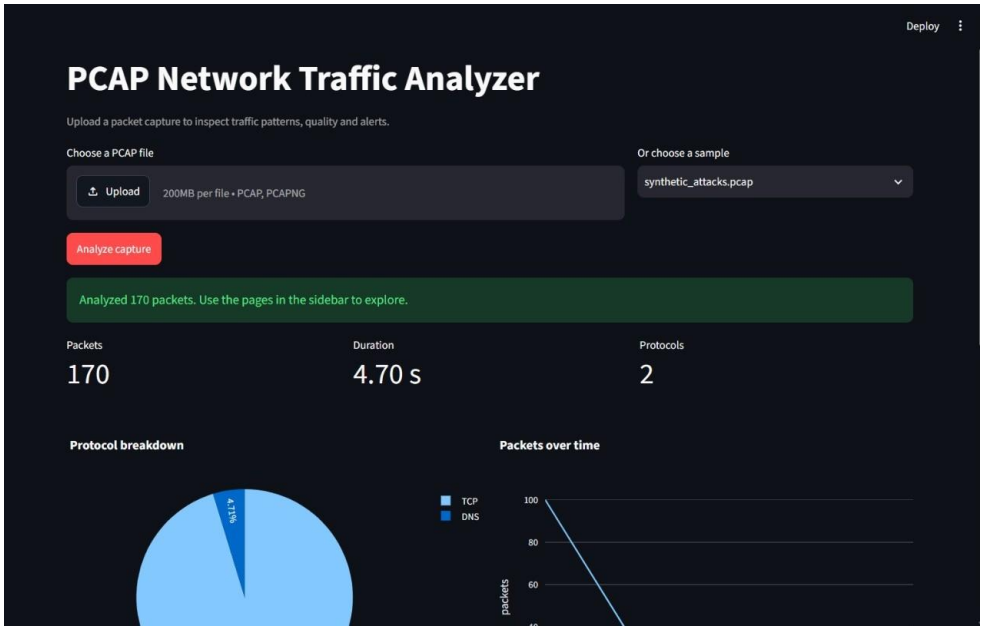


Figure 6.1: Home page of the PCAP Network Traffic Analyzer (output screenshot). The sample synthetic_attacks.pcap has been analysed: 170 packets, 4.70s and 2 protocols.

6.2 Protocol Mix and Packet Rate

Figure 6.2 shows the protocol breakdown and the packets per second. TCP makes up 95.3% of the packets and DNS 4.71%, that is 162 TCP and 8 DNS packets. About 100 packets arrive in the first second and the rate then falls to about 30 per second and below, so most of the attack traffic is sent at the start of the capture.

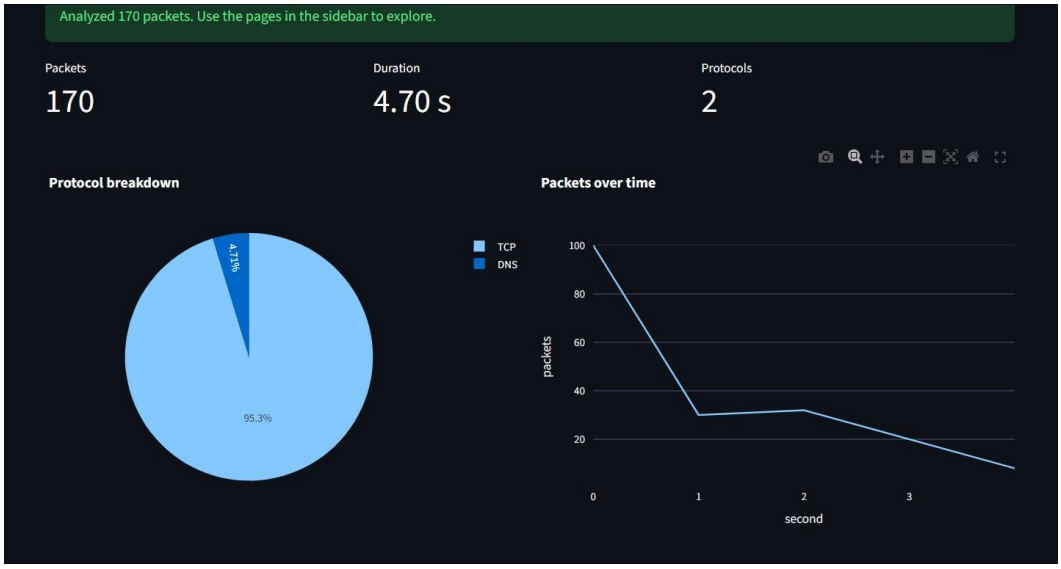


Figure 6.2: Protocol breakdown and packets over time (output screenshot). TCP is 95.3% of the packets and DNS 4.71%; the packet rate peaks at about 100 packets in the first second and then falls.

6.3 TCP Handshake Analysis

Table 6.1 and Figure 6.3 give the TCP results. The application found 162 TCP flows, every one of them half-open, and no retransmissions or duplicate ACKs. Each TCP packet in the capture therefore opens its own flow and is never answered, which is the pattern a SYN flood and a SYN port scan both produce.

Table 6.1: TCP analysis of the synthetic capture.

Measure	Value
TCP flows	162
Half-open flows	162 (100%)
Completed handshakes	0
Retransmissions	0
Duplicate ACKs	0



Figure 6.3: TCP analysis page (output screenshot): 162 TCP flows, no retransmissions or duplicate ACKs, and all flows in the half-open state.

6.4 Alerts

Figure 6.4 shows the Alerts page, and Tables 6.2 and 6.3 list its contents. All three attacking hosts were flagged, each with the correct attack and a reason in plain words. 10.0.0.7 is the only High alert, with a risk score of 100; the SYN flood source 10.0.0.5 and the port scanner 10.0.0.9 are Medium.

Table 6.2: Alerts per source IP.

Source IP	Severity	Risk score	ML score	Reason
10.0.0.5	Medium	70	50.9	Sent 130 SYNs with a low SYN-ACK ratio in a 10-second window
10.0.0.7	High	100	100	Long or random-looking DNS labels were queried
10.0.0.9	Medium	45	0	Scanned 32 destination ports in a 10-second window

Table 6.3: Rule findings.

Rule	Source IP	Window	Severity points
SYN flood	10.0.0.5	2000	70

DNS tunnelling	10.0.0.7	2000	55
Port scan	10.0.0.9	2000	45

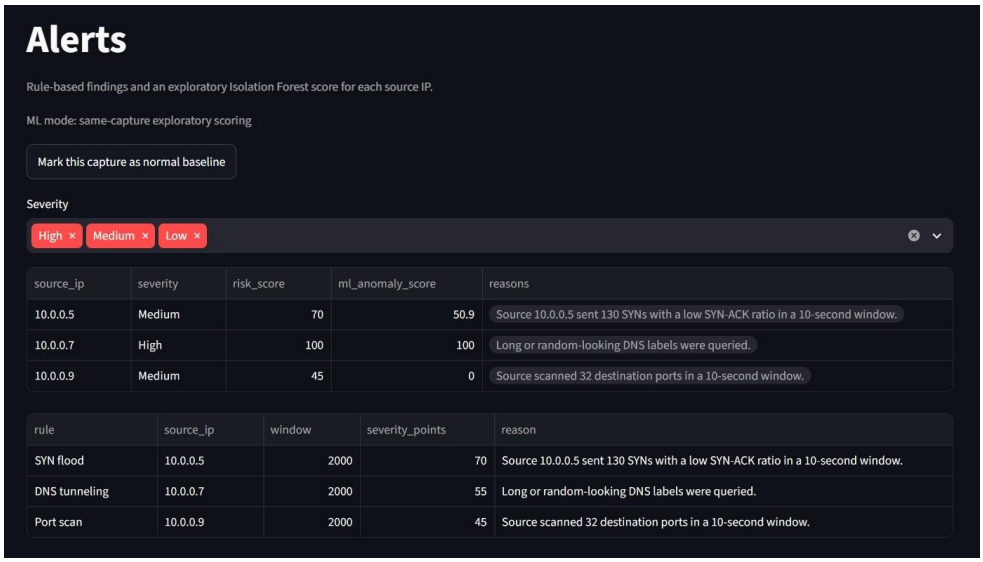


Figure 6.4: Alerts page (output screenshot): one row per flagged source with severity, risk score, Isolation Forest score and reason, followed by the individual rule findings.

6.5 Top Talkers

Table 6.4 and Figure 6.5 list the hosts that sent the most traffic. Dividing bytes by packets gives 54 bytes for every packet of 10.0.0.5 and 10.0.0.9, which is exactly a 14-byte Ethernet header, a 20-byte IP header and a 20-byte TCP header with no data: bare SYN packets, as expected for a SYN flood and a SYN scan. The DNS queries of 10.0.0.7 average 121 bytes, larger than a typical short query because of their long labels.

Table 6.4: Top talkers.

Source IP	Packets	Bytes	Bytes per packet
10.0.0.5	130	7,020	54
10.0.0.9	32	1,728	54
10.0.0.7	8	968	121
Total	170	9,716	57.2

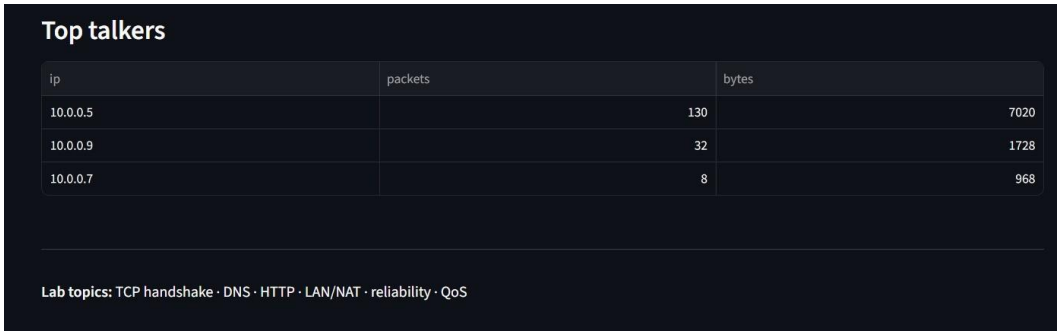


Figure 6.5: Top talkers table on the home page (output screenshot). 10.0.0.5 sent 130 of the 170 packets.

6.6 Rules versus Isolation Forest

The rules flagged all three attacking hosts, a detection rate of 100% (Equation 4.5) on this capture, with no false alerts. The Isolation Forest gave high scores to 10.0.0.7 (100) and 10.0.0.5 (50.9) but a score of 0 to the port scanner 10.0.0.9. With only three hosts, all of them attackers, “anomalous” can only mean “different from the other two”; the port scanner sits between the other two hosts in packet count and bytes, so the forest does not isolate it quickly. The application itself labels this mode as exploratory, and the result shows why the rule findings, not the anomaly score, should drive the alerts on such a capture.

CHAPTER 7

DISCUSSION

7.1 Key Findings

- All three injected attacks were detected and explained. The SYN flood, the port scan and the DNS tunnelling each produced one alert for the right host, with a reason that names the evidence (130 SYNs, 32 ports, long DNS labels).
- The traffic statistics agree with each other. 162 TCP and 8 DNS packets make the 170 packets; 162 half-open flows match the 130 + 32 SYN packets; and 54 bytes per TCP packet is exactly the size of a bare SYN.
- The anomaly score is not reliable on a capture with no normal traffic. The Isolation Forest missed the port scanner, which the rule caught. The score is useful as a second opinion next to the rules, not on its own.
- The overview alone already points to an attack. A burst of about 100 packets in the first second, almost all TCP and all half-open, is visible before the Alerts page is opened.

7.2 Limitations

- All results come from one synthetic capture of 170 packets; detection rate and false alerts on real traffic have not been measured.
- The capture contains no normal traffic, so the false-alarm behaviour of the rules on busy, legitimate traffic is unknown.
- Thresholds of the rules were set by hand and have not been tuned or validated on labelled data.
- The DNS, HTTP, LAN and QoS pages were not exercised in the results shown here; the sample has no HTTP traffic.
- The application analyses saved captures only, not live traffic.

7.3 Future Work

The next step is to test the analyzer on public captures with labelled attacks and on normal campus traffic, and to report precision and recall for each rule. The Isolation Forest should be trained on a baseline of normal traffic, using the existing baseline option, and compared with the same-capture mode.

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